East West University

Department of Computer Science and Engineering

Summer 2022

Lab Assignment Submission Report-03

Course Code: CSE302

Course Title: Database Systems

Section: 02

SUBMITTED BY

Student Name: Jannatul Ferdous Salma

Student ID: 2020-2-60-005

SUBMITTED TO

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Associate Professor,

Department of Computer Science and Engineering

DATE OF SUBMISSION

22 JUNE 2022

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Name: Jannatul Ferdous Salma
ID: 2020-2-60-005
Sec: 02

Relational Algebra

(1)

$$\pi_{\text{Patient_id, Patient_name, Patient_age, Patient_contact}} \left(\sigma_{\text{Patient.Patient_id = visit.Patient_id and doctor.doctor_id = visit.doctor_id and speciality = 'Heart'}} \left(\begin{array}{c} \text{Patient} * \\ \text{doctor} * \\ \text{visit} \end{array} \right) \right)$$

(2)

$$\pi_{\text{Patient_id, Patient_name}} \left(\sigma_{\text{Patient.Patient_id = visit.Patient_id and doctor.doctor_id = visit.doctor_id and doctor_name = 'Alice Brown' and visit_date < '06-20-2022''}} \left(\begin{array}{c} \text{Patient} * \\ \text{doctor} * \\ \text{visit} \end{array} \right) \right)$$

(3)

$$\pi_{\text{Prescription_id, general finding, advice, visit_id}} \left(\sigma_{\text{patient.Patient_id = visit.patient_id and Prescription.visit_id = visit.visit_id and Patient_age > 25}} \left(\begin{array}{c} \text{Patient} * \\ \text{visit} * \\ \text{Prescription} * \end{array} \right) \right)$$

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(4)

$$\pi_{\text{Patient_id, Patient_name, Patient_age, Patient_contact}} \left(\left(\sigma_{\text{Patient.Patient_id = visit.patient_id and visit.doctor_id = doctor.doctor_id and doctor_name = 'Alice Brown'}} \left(\begin{array}{c} \text{doctor} * \\ \text{Patient} * \\ \text{visit} \end{array} \right) \right) \right)$$

$$\cup$$

$$\left(\sigma_{\text{Patient.Patient_id = visit.Patient_id and visit.doctor_id = doctor.doctor_id and doctor_name = 'Robin Steve'}} \left(\begin{array}{c} \text{doctor} * \\ \text{Patient} * \\ \text{visit} \end{array} \right) \right)$$

$$-$$

$$\left(\sigma_{\text{Patient.Patient_id = visit.Patient_id and visit.doctor_id = doctor.doctor_id and doctor_name = 'Robin Steve'}} \left(\begin{array}{c} \text{doctor} * \\ \text{Patient} * \\ \text{visit} \end{array} \right) \right)$$

(5)

$$\pi_{\text{doctor_id, doctor_name, specialty}} \left(\left(\sigma_{\text{Patient.Patient_id = Visit.Patient_id and doctor.doctor_id = Visit.doctor_id and visi_date <= '06-20-2022'}} \left(\begin{matrix} \text{doctor} * \\ \text{Patient} * \\ \text{visit} \end{matrix} \right) \right) \right)$$

$$\cup$$

$$\left(\sigma_{\text{Patient.Patient_id = Visit.Patient_id and doctor.doctor_id = Visit.doctor_id and visit_date > '06-20-2022'}} \left(\begin{matrix} \text{doctor} * \\ \text{Patient} * \\ \text{visit} \end{matrix} \right) \right)$$

$$-$$

$$\left(\sigma_{\text{Patient.Patient_id = Visit.Patient_id and doctor.doctor_id = visit.doctor_id and visit_date = '06-20-2022'}} \left(\begin{matrix} \text{doctor} * \\ \text{Patient} * \\ \text{visit} \end{matrix} \right) \right)$$

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